The Paleomagnetism of Seal and Croteau Rocks from the Grenville Front, Labrador: Polar Wandering and Tectonic Implications[1]

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Some of the red beds of the Seal Group contain at least two CRM (chemical remanent magnetization) components. The dominant one has a direction of 275°, +05° (pole position 205° E, 05° N). The less pronounced component has a direction which is about 55° clockwise from this. It is suggested that the dominant (westerly) component is the younger one and was developed after the block containing these rocks was rotated clockwise.

The igneous rocks of the Seal and Croteau Groups are reversely magnetized relative to the red beds and yield a pole (152° E, 05° N) about 50° away from that indicated by the major component of the red beds. The magnetization of the igneous rocks is thought to contain a primary TRM (thermoremanent magnetization) but the igneous pole is 65° of arc away from poles derived from rocks of similar age in the Superior Province. If, however, a correction of 55° rotation (as suggested by the red-bed results) is applied to the block, the igneous pole then lies in the southern hemisphere (197° E, 22° S) close to poles for rocks of similar age from the Superior Orogen. It is then suggested that the primary magnetization of the igneous rocks was acquired before rotation. Because of its pole position, it appears that the dominant CRM (post-rotation) was acquired less than 400 m.y. after the initial magnetization of the igneous rocks. This places an upper limit to the interval of time during which the rotation occurred. It is suggested that the rotation is perhaps related to early Grenvillian orogenic and thermal events (1200–1500 m.y.).

Paleomagnetic evidence when coupled with geological evidence suggests that not only was the N.E. Grenville Orogen rotated, but that in the interval 1550–1200 m.y. ago, the paleopole relative to this part of the crust (and to the Superior) underwent an excursion, referred to as the "Grenville Loop", deep into the Southern hemisphere.

Bien des grès rouges du groupe Seal contiennent (au moins) deux composantes d'aimantation rémanente cristalline (CRM). La direction de la composante dominante est 275°, +05° (position du pôle: 205° E, 05° N) tandis que la direction de l'autre composante est environ 55° plus au nord (≃330°). Nous suggérons que la composante dominante est la plus récente des deux et qu'elle fut acquise après une rotation dextrorsum de la couche terrestre contenant ces grès.

La direction (142°, +42°) de l'aimantation principale, que l'on retrouve dans les roches ignées des groupes Seal et Croteau, donne un pôle (152° E, 05° N) qui est situé à 50° du pôle des grès. Nous croyons que cette aimantation est une aimantation thermorémanente (TRM) primaire quoique le pôle soit éloigné de 65° d'arc de certains autres pôles obtenus de roches, d'à peu près le même âge, qui proviennent de la province du Supérieur. En effet, si l'on applique une rotation sinistrorsum de 55° (selon les résultats des grès) à la dite couche terrestre, on retrouve le pôle des roches ignées dans l'hémisphère sud (197° E, 22° S) beaucoup plus près de ces autres pôles suggérant que l'aimantation thermorémanente des roches ignées eut lieu avant la rotation. D'après la position de son pôle, il semble que la composante dominante de l'aimantation rémanente cristalline fut acquise en deçà de 400 m.a. de l'aimantation des roches ignées, ce qui permet d'établir la limite de temps pendant lequel la rotation en question se produisit. Il est suggéré que cette rotation eut peut-être lieu au début des événements orogéniques et hydrothermiques du Grenvillien (1200–1500 m.a.).

Quand les résultats paléomagnétiques sont étudiés à la lumière des observations géologiques, il apparait que, pendant l'intervalle 1550–1200 m.a. passées, (1) une bonne partie du nord-est de l'orogenèse de Grenville fut soumise à une rotation, et (2), relativement à cette croûte terrestre (et à celle du Supérieur), le paléopôle entreprit un voyage prolongé, décrit par la "Boucle de Grenville", dans l'hémisphère sud.

Introduction

The nature of the Grenville Front has been widely discussed in recent years (Stockwell 1964, Grasty *et al.* 1969, Stevenson 1970, Taylor 1971, Fahrig and Larochelle 1972, Irving *et al.* 1972*c*). The Croteau and Seal

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Groups lie near the Front and because of their location, were chosen for paleomagnetic work. Both red beds and igneous rocks that are suitable for paleomagnetic work are available in the Seal Group and study of both is desirable as the two types of rocks acquire their magnetization by different processes. The rocks of the Seal and Croteau were formed in the interval ($\simeq$1200 to $\simeq$1500 m.y. ago) during which time the paleomagnetic pole (relative to the Precambrian Shield of North America) was previously thought to have remained almost stationary near the present-day equator (Palmer 1969, Spall 1971).

Geology

The Croteau and Seal Groups form an elongate belt of volcanic and sedimentary rocks that is preserved chiefly adjacent to, and north of the Grenville Front in Labrador (Fig. 1). The Croteau as defined (Fahrig 1959) consists of two parts, the lower being made up of black pyritic shales, graywacke, quartzite, dolomite, and metabasalt. This is overlain unconformably along Croteau and Pocket Knife Lakes by a basal conglomerate followed by sandstones. The latter is feldspathic and consists of alternating pink, maroon, and cream colored layers. Coarser phases of the sandstone are cross bedded and locally have a carbonate cement. The sandstones may be partly tuffaceous as devitrified shards are locally abundant. Above this basal zone the Croteau Group consists largely of intermediate to acid, purplish to green volcanic porphyries with thin layers of cross-bedded, tuffaceous cream to purple sandstone.

The above mentioned unconformity appears, however, to be of major significance so it is more reasonable to provide separate group names for the upper and lower sequences. In the present report the name Croteau will be restricted to the predominantly volcanic sequence that lies above the unconformity. The Croteau Group thus defined, is Helikian in age, whereas the rocks below the unconformity may be significantly older, perhaps Aphebian in age.

In the Snegamook Lake map area (Fahrig 1957, 1959, Brummer and Mann 1961) the strataform rocks of the Seal Group fall into five major lithostratigraphic units. The predominant rock types of the five units, proceeding from the base of the group upward, are: (1) shales and coarse clastics, (2) volcanics and sandstones, (3) diabase with interlayered sandstone and shale, (4) volcanics and shale, and (5) sandstone. The presence of two separate volcanic sequences suggests cyclicity of igneous activity; however, the sedimentary rocks between the two volcanic sequences are extensively intruded by gabbro and diabase sills. It is conceivable that these intrusions are evidence of continuing igneous activity in the interval between the two periods of volcanism. The sedimentary rocks of the Seal Group exhibit an oscillatory but steady upward fining from the base of the group to the base of the uppermost preserved unit which is sandstone. The termination of the upward fining and the appearance of the upper sandstone coincides approximately with the end of the second period of volcanism.

Volcanic porphyries similar to those of the Croteau are present near the base of the Seal Group in an area about 130 km to the west of the Croteau rocks. Regional structural relations of the Seal and Croteau Groups in the Snegamook Lake map area (Fahrig 1957) and the relationship of Croteau-type porphyries to Seal Group rocks alluded to above suggest that the deposition of the Croteau Group either predated or was early relative to Seal Group deposition; however, deposition of the two sequences may have been locally continuous.

The lithology of the Seal Group suggests that its depositional environment was chiefly continental grading at times into shallow marine. The volcanics of the Seal Group are amygdaloidal and intensely reddened at the top and base of flows as well as along shrinkage joints. The coarse sedimentary rocks of the basal unit are immature and many of the shaly units are red, ripple marked, and exhibit desiccation cracks. The lower Croteau sediments are also immature and some of the tuffaceous sandstones interbedded with volcanics higher in the sequence are intensely cross bedded and reddened as are some of the Croteau volcanic rocks. The Croteau may also then be predominantly continental.

The relationship of the Croteau and Seal Groups to the great Harp Lake anorthosite intrusion which outcrops just to the north of

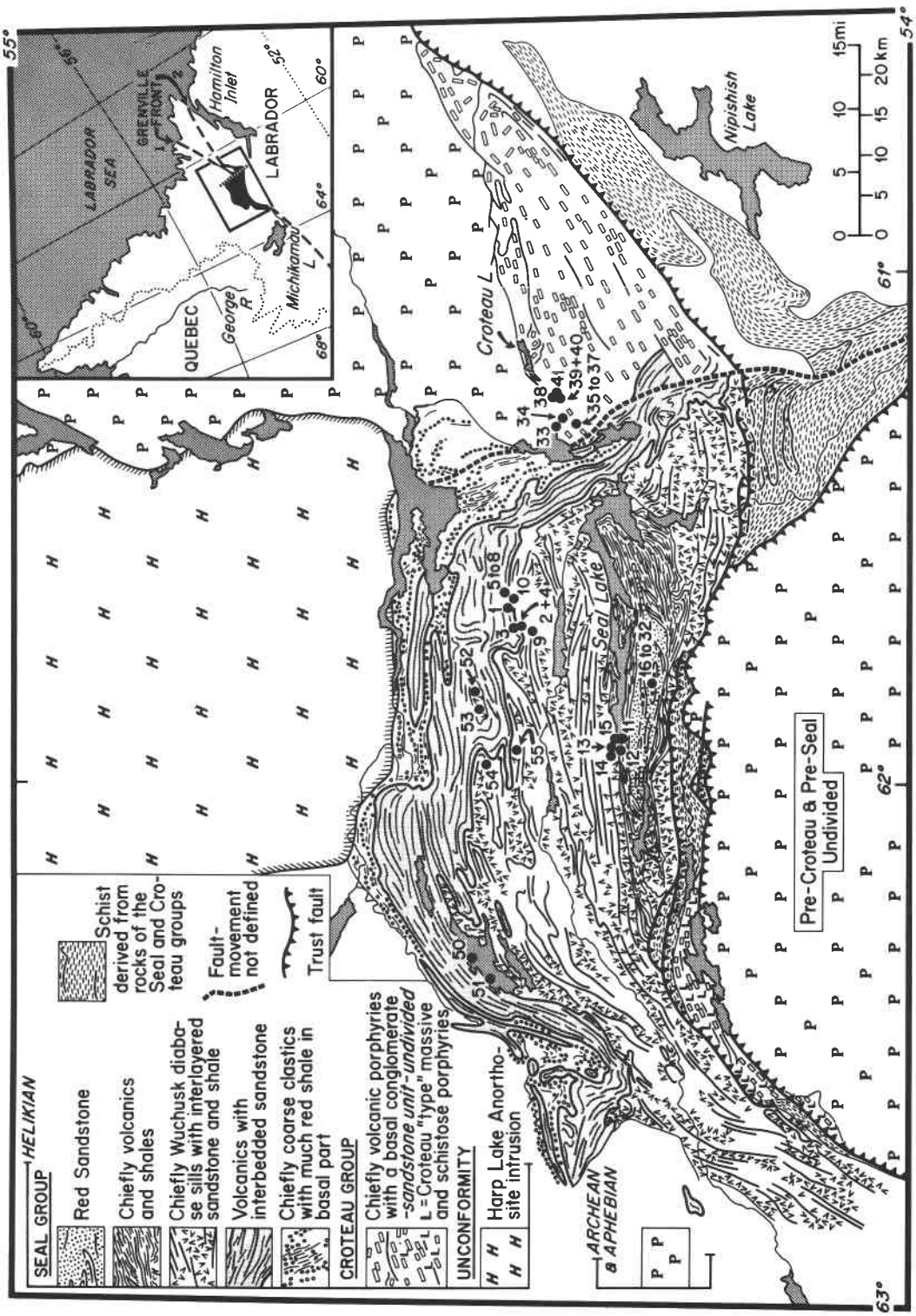


FIG. 1. Locality map showing the sampling sites. The Grenville Front according to: (1) Tectonic map of Canada (1251A) and (2) Taylor (1971).

Seal Group strata is of particular interest in terms of geochronology and paleomagnetism. The contact between the Seal Group and Harp Lake intrusion has not been observed; however, lower Seal strata appear to overlap gently on the anorthosite. This suggests that the Seal Group lies unconformably on the intrusion, a conclusion supported by aeromagnetic maps of the region. On these maps the Harp Lake magnetic pattern may be traced southward under the Seal Group to Wuchusk Lake, a distance of 19 km. Finally, R. F. Emslie (personal communication 1972) has observed red clastic "dikes" cutting the anorthosite. These may be Seal Group equivalents.

The most reliable radioisotopic datum relating to the age of the Croteau and Seal Groups consists of a Rb/Sr isochron on Croteau volcanics. The age obtained is 1563±43 m.y. using $\lambda = 1.39 \times 10^{-11}$ yr^{-1} (Wanless and Loveridge 1972, p. 57). If, as suggested above, the Croteau and Seal Groups represent essentially continuous continental deposition, this age may be considered a rough estimate of the age of both groups. It also indicates that the Harp Lake anorthosite, unconformably below the Seal, has an age greater than 1563±43 m.y.

The presence of a great thickness of continental volcanic and sedimentary rocks of essentially molasse facies suggests deposition of the Seal strata in a graben structure. Early rapid subsidence with pronounced local relief is suggested by the coarse, immature basal clastics. The upward fining of the sedimentary rocks suggests that the rate of filling of the basin by both igneous and sedimentary rocks was rapid enough relative to any subsequent subsidence to produce a gradual reduction in relief at least until the deposition of the uppermost preserved unit. The alkalic nature of some of the igneous rocks is quite characteristic of volcanic activity in some of the world's younger rift valley systems.

Both the Croteau and Seal rocks were folded subsequent to any deformation that may have occurred during the filling of the Croteau–Seal rift (graben). The Croteau rocks now form a northeasterly trending system of folds and the fold axes in the more intensely deformed areas of Seal rocks now describe an arc, open to the south. The arc probably resulted, at least, in part from thrusting directed from the south, as the southern limit of the fold arc consists of major thrust faults along which basement gneisses have been pushed north over rocks of the Seal Group. The sequence of deformational events in this region may be complex. Some light is thrown on the subject by the present paleomagnetic study so further discussion has been reserved for the final sections of this paper.

Samples of Seal rocks for paleomagnetic study were obtained in a section extending from the upper part of the lower volcanics (see sites 5 to 8 of Fig. 1) to the uppermost preserved unit which is red sandstone (sites 16 to 32 of Fig. 1). If the Brummer and Mann (1961) estimate of 10 260 m for the thickness of the Seal Group is reliable, the samples represent about 9000 m of section. The Croteau volcanic samples are from a gently southeasterly-dipping monoclinal sequence and probably represent 600 to 1200 m of flows.

Paleomagnetic Study

The magnetization of samples from 46 sites of the Seal and Croteau Groups was studied. The pole positions obtained from the sedimentary and igneous rocks differ by some 50°. Furthermore, the cleaning techniques and the methods used to separate the different magnetizations contained in the sedimentary and the igneous rocks were different. For these reasons the sedimentary and igneous rocks results are described separately.

Sedimentary Rocks (Seal Group Only)

Three to five blocks were collected from each site, except for sites 4 and 10 where core samples were drilled in the field. One to three cylindrical specimens (2.5 cm in diameter and 2.2 cm in height) were taken from each block; the number of specimens (N) thus obtained is given in Table 1. The intensity (M_i—Table 1) of the natural remanent magnetization (NRM) was weak (in some specimens, it was $<1 \times 10^{-6}$ e.m.u.) and tests showed that many specimens could easily acquire a magnetization in the earth's field. However, this magnetization can be removed by thermal demagnetization at 100 °C.

All specimens were found to carry a thermally stable magnetization with blocking temperatures above 650 °C and usually up to the

TABLE 1. Analysis of intensities and directions of sediments before (M_N) and after (M_T) thermal cleaning at temperature T*

			M_N(NRM)				M_T (cleaned)				
Site	N	n	M_i	$D°, I°$	R	$\delta°$	M_i	$D°, I°$	R	K	$\delta°$
					Group 1						
4	8	2†	1.4	134,+36	1.800	26	0.1	277,+03	1.969	33	10
10	5	4	2.9	231,+45	3.315	34	1.8	271,+15	3.669	10	22
16	4	3	2.0	261,+47	2.649	28	1.5	276,−01	2.847	16	18
18	7	4	2.4	229,+22	2.697	48	1.4	268,+07	3.849	20	16
19	7	6	2.9	261,+19	5.374	26	1.5	269,+05	5.715	18	18
21	6	2†	3.6	332,+16	1.983	8	1.3	297,+09	1.878	8	19
22	5	5	28.3	293,+09	4.973	6	7.6	288,+09	4.940	66	9
23	5	3	5.4	318,+16	2.759	23	2.6	283,+08	2.963	53	9
24	5	2†	2.9	296,+30	1.979	9	0.8	273,+01	1.972	35	10
25	6	6	3.5	277,+33	5.023	33	2.0	278,+05	5.861	36	12
26	7	6	2.5	297,+46	5.365	27	1.5	278,+13	5.773	22	16
27	4	3	1.1	276,+05	1.084	69	0.9	267,−15	2.970	67	8
29	8	5	4.5	264,+24	4.320	30	3.1	275,+05	4.952	83	8
32	5	2†	0.7	155,+20	1.786	27	0.6	263,+07	1.973	37	9
Mean (a)‡	14							276,+05	13.744	51	11
Mean (b)‡	10†							275,+05	9.843	57	10
					Group 2						
17	6	1	1.9	137,+06	—	—	1.0	313,+22	—	—	—
20	4	4	4.7	012,+24	3.684	23	1.1	324,+12	3.204	4	37
28	6	6	1.7	196,+24	4.317	47	0.4	233,−10	4.339	3	44
30	5	0	—	—	—	—	—	—	—	—	—
31	8	2	1.6	302,+84	1.020	59	0.7	309,+19	0.865	1	64

NOTES: N = number of specimens measured; n = number of specimens where $\Delta\Psi$ (index of within-specimen uniformity of direction) < 20°; M_i = intensity × 10^{-6} e.m.u. (CGS system); R = resultant vector; k = precision estimate (Fisher 1953); $\delta^0 = \cos^{-1}(R/n)$ = angular standard deviation (Wilson 1959).

*Directions obtained after demagnetization in zero field at temperatures between 650 and 685 °C—see text and Figs. 2 and 4 for explanation of the 650–685 °C cleaning temperatures.

†Excluding sites where $n < 3$.

‡For (a) $\alpha_{95} = 5.6$; pole position 205° E, 06° N (3, 6). For (b) $\alpha_{95} = 6.4$; pole position 205° E, 05° N (3, 6). Note that (3, 6) are semi-axes (δp, δm) of the ellipse of confidence at $P = 0.05$ (Irving 1964).

Curie point of hematite. In most specimens, the magnetization was not homogeneous, that is, the magnetization direction was not the same throughout the specimen but was differently directed in different parts of the specimen. In many instances, all but one uniformly directed magnetization could be removed from the specimen. The method by which this composite magnetization was detected is described first.

The instrument used for measuring was an astatic magnetometer with negative feedback (Roy *et al.* 1972*a*). The magnetometer is operated as a null detector and the direction of magnetization is given directly. The direction (D_1) of the horizontal component of magnetization (with respect to a reference mark) is obtained by placing the specimen in an upright position and directly under the instrument; the specimen is then rotated until a zero deflection indicates the direction of magnetization which is read on a 0–360° graduated circle. The specimen is then inverted and the procedure repeated to obtain D_2. The inclination (I_1) of magnetization with respect to the axis of the specimen is obtained by placing the specimen on its side with the horizontal component in the horizontal plane; the specimen is then rotated 180° about its axis and the inclination remeasured to obtain I_2. This method is quite effective in providing an estimate of the uniformity of direction of magnetization within the specimen. When the magnetization is unidirectional $D_1 = D_2$ and $I_1 = I_2$. When $D_1 \neq D_2$, the upper and lower parts of the specimen are not magnetized in the same direction. When $I_1 \neq I_2$, the inclination is not linear. Comparison of D_1 with D_2 and I_1 with I_2 then provides a means of estimating the reliability (for paleomagnetic purposes) of the mean direction of magnetization of a single specimen. Large inequalities between D_1 and D_2 and/or I_1 and I_2 indicate that the mean direction and

the mean intensity represent the resultant vector of two magnetizations acquired along different directions. Such inequalities appear to result from changing conditions (field direction changes or tectonic activity) *during* magnetization and not from the superimposition of two magnetizations of clearly different origins like say, a TRM and VRM (viscous remanent magnetization) acquired at different times in two fields, each with a different but (essentially) unvarying direction. In the latter case, one would expect each magnetization to be uniformly directed. Assuming that the magnetic particles are homogeneously distributed, the vector sum of the two magnetizations will also be uniformly directed. A few experiments were performed on some of the specimens whose magnetization was uniformly directed. They were placed for a week in the earth's field in such a position that the direction of a viscous magnetization would differ greatly from that of the NRM. Although the direction changed by as much as 30° from that of the NRM, the magnetization remained uniformly directed. The specimens were then thermally demagnetized at 100 °C in zero field. The direction of magnetization changed again but remained unidirectional throughout the specimen. So, two magnetizations acquired in different but uniformly directed fields have not, in this case, produced a non-uniformly directed magnetization. It is suggested that the most probable cause of non-uniformity of direction within a specimen is that, *during* magnetization, the direction of the field changed relative to the attitude of the specimen. This, of course, implies that the magnetization of sediments is acquired over a period (Roy and Robertson 1968) long enough to record field direction changes with respect to the specimen.

Whatever the cause of non-uniformity, it is evident that if the direction varies greatly within the specimen, the result is not a reliable indicator for calculating pole positions and should be eliminated from such calculations. Different criteria have been used for estimating the reliability of observations. For example, at the site level, the randomness test (Watson 1956) and the angular standard deviation test (Wilson 1959) are commonly used to estimate the reliability of the site mean direction; at the core level, Larochelle (1969) has used the angular difference between the directions obtained from two consecutive specimens of a core sample to estimate the reliability of the core mean direction. Analysis of directions at the site or core level are applicable only if the magnetization was acquired over a relatively short period of time (*e.g.* during cooling of lavas) so that the mean direction may be taken to represent the field at a single point in time. The Seal Group sediments, however, appear to have acquired their magnetization over a considerably longer period so that specimens (or even different parts of a specimen) from a given site or core may have acquired their magnetization at significantly different times. In this instance, analysis at the site or core level is not adequate. It is then much preferable to perform an analysis of directions at the specimen or within-specimen level. With the direct reading method described above, it can be readily determined if the magnetization carried by a specimen was acquired along a single direction or in different directions, that is whether or not the *mean* direction can be associated with a particular field. A new criterion (within-specimen uniformity of direction) for estimating the reliability of direction at the specimen level is described in the following section.

Thermal Demagnetization

One or two specimens from each site were subjected to thermal demagnetization using a non-magnetic oven (Roy *et al.* 1972*b*) in a quasi-zero field (Roy *et al.* 1969). In general, specimens whose magnetization directions were least uniform (largest difference between D_1 and D_2 and/or I_1 and I_2) were chosen.

In the range 20 to 650 °C, there was little change in direction except when the NRM direction was far away from the final direction of 275°, +05° (Figs. 2 and 4); in these instances, after heating to successively higher temperatures, the directions moved progressively towards 275°, +05°. However, whether the direction changed or not, the intensity did not decrease appreciably (Fig. 3) and there was little increase in the degree of within-specimen uniformity of direction in that temperature range indicating the presence of two thermally stable but differently directed magnetizations which may or may not be of the same origin.

In the 650 to 685 °C range, the within-

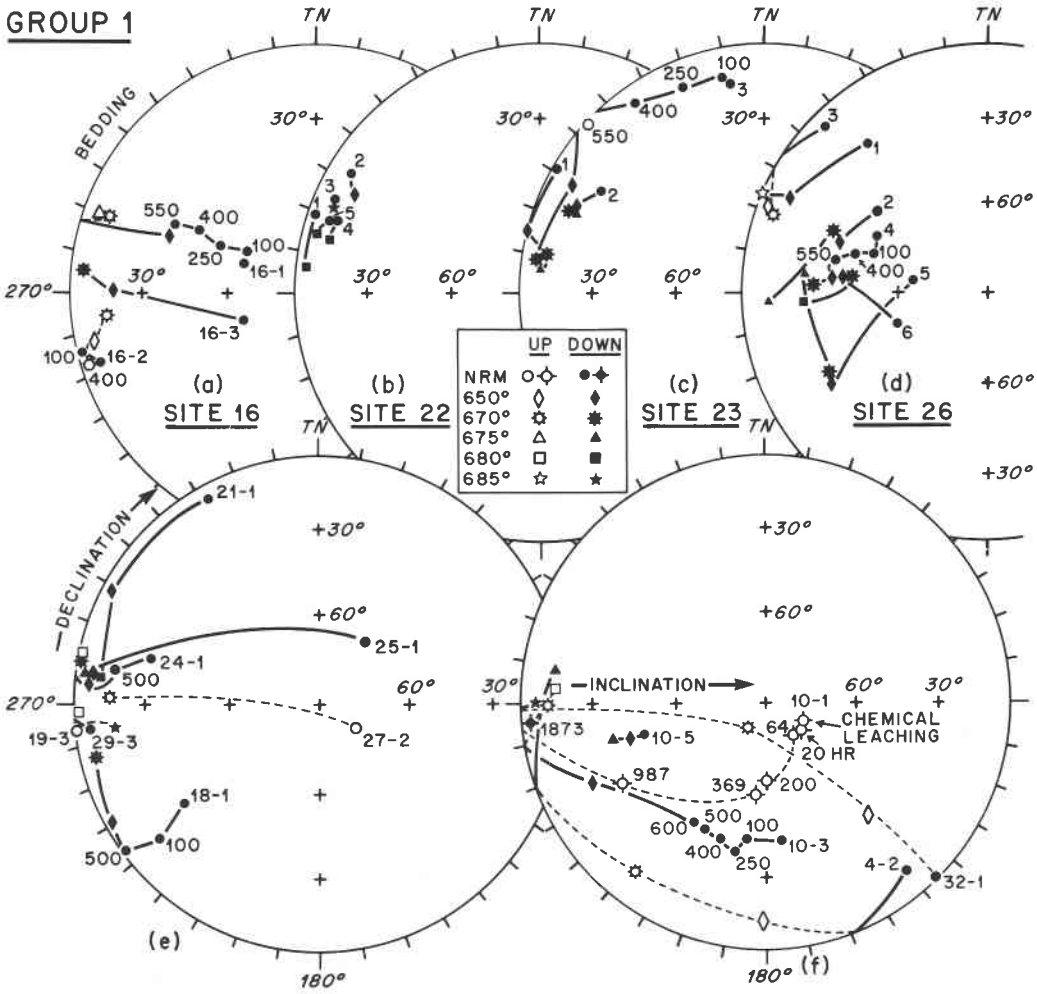


FIG. 2. Directions of magnetization before and after thermal demagnetization at the temperature shown. The directions obtained after demagnetization at temperatures of 650 °C or higher are indicated by the appropriate symbols. Equal area polar projection. Closed (open) symbols indicate positive (negative) inclinations. The last point of each curve indicates the cleaning temperature after which $\Delta\Psi$ was not only $\leqslant 20°$ but also the smallest of the $\Delta\Psi$ obtained after each demagnetization steps between 650 °C and 685 °C. (a), (b), (c), and (d) show the directions of all specimens from 4 sites of Group 1 (Table 1); (e) and (f) show the directions of one specimen from each of the other sites included in Group 1; and (f) also shows the directions obtained after successive immersions in hydrochloric acid during the number of hours shown. The Group 1 results are used to calculate the pole position.

specimen uniformity of direction was sometimes greatly increased, indicating that one of the magnetizations was, at least, greatly reduced. Since this increase did not occur at a specific temperature but at different temperatures for different specimens, the remaining specimens were demagnetized successively at the following temperatures: 650, 670, 675, 680, and 685 °C or until they reached an intensity lower than 0.1×10^{-6} e.m.u. ($M_i < 10^{-7}$ e.m.u.). In some instances, one or two steps before reaching $M_i < 10^{-7}$ e.m.u., the uniformity was greatly increased; in other instances, it remained low. It therefore appears that the blocking temperature spectra of the two magnetizations are close to each other and possibly overlap, making it difficult, if not impossible, to separate them in every case. It is reasonable to assume

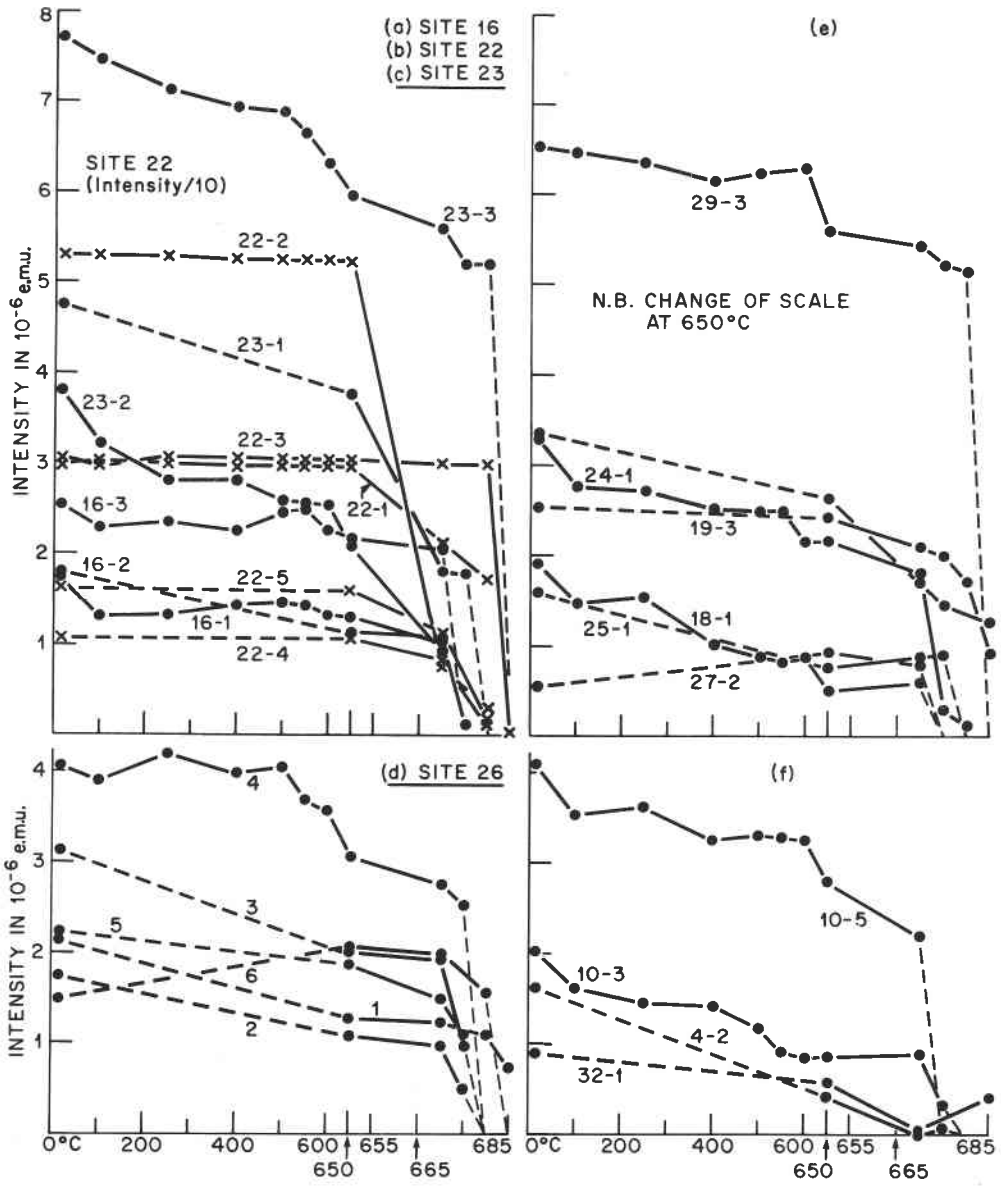


FIG. 3. Intensity decay curves of all specimens whose directions are shown in Fig. 2. The dashed line towards the abscissa indicates that the intensity at that temperature was less than 1×10^{-7} e.m.u. Note that these curves show the complete demagnetization curves and the smallest $\Delta\Psi$ (indicated by the last point in Fig. 2) was not necessarily obtained on the last demagnetization step shown here. In some instances, the $\Delta\Psi$ increased slightly as the demagnetization temperature was taken one step higher although the mean direction did not change appreciably. This is attributed to a small temporary (viscous) magnetization acquired in the field of the magnetometer; the effect of which becomes more evident as the intensity of the stable magnetization is further reduced after demagnetization at a higher temperature.

that where the uniformity is high, the two magnetizations have been separated by eliminating or at least greatly reducing one of them so that the remaining magnetization can be used to estimate the field direction (assuming a dipole field) relative to these rocks at the time of magnetization. If the uniformity remains low, the two magnetizations have not been separated

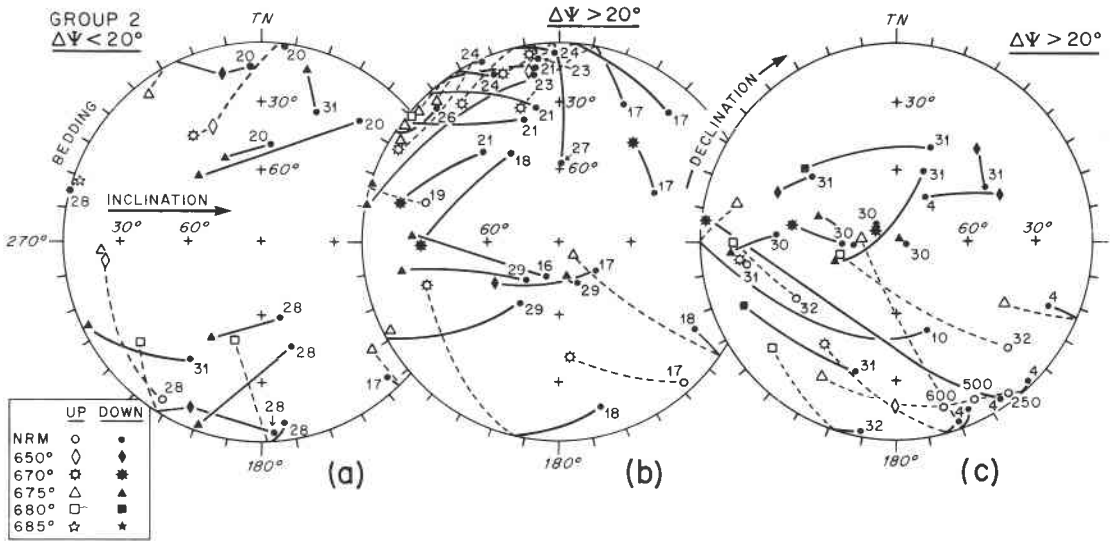


FIG. 4. Symbols and projection as per Fig. 2(a) shows the directions of all specimens with a $\Delta\Psi \leqslant 20°$ but where the within-site dispersion $\delta \geqslant 37°$ (Group 2). (b) and (c) show the direction of *all* specimens where $\Delta\Psi$ was constantly >20°.

and consequently the direction of the resultant vector is not parallel to the direction of a single magnetizing field. The mean direction of magnetization of that specimen cannot then be used to determine directly the direction of the earth's field although the within-specimen variation of direction may provide, as seen later, some information about the magnetization history.

To determine whether one magnetization has been isolated or not, we have used an index ($\Delta\Psi$) of within-specimen uniformity of direction using the empirical formula

$$\Delta\Psi = \left[\frac{(\Delta D \cos I)^2 + \Delta I^2}{2}\right]^{1/2}$$

where $\Delta D = D_1 - D_2$, $\Delta I = I_1 - I_2$, and $I =$ mean inclination with respect to the specimen. Since $\Delta\Psi$ is dependent upon the distance (z) between the specimen and the lower magnet, all measurements were made at the same distance ($z = 4.08$ cm). $\Delta\Psi$ was then calculated for all 111 specimens and for each demagnetization step. Taking the lowest $\Delta\Psi$ obtained after cleaning at 650 °C or higher, the data were divided into two categories: $\Delta\Psi \leqslant 20°$ and $\Delta\Psi > 20°$. The 20° limit was chosen according to the $\Delta\Psi$ distribution shown in Fig. 5. The direction changes for most of the specimens are given in Figs. 2 and 4. The last point of each curve indicates the temperature at which the smallest $\Delta\Psi$ occurs. As indicated by n in Table 1, 66 specimens fall into the category $\Delta\Psi \leqslant 20°$ which is further subdivided into Group 1 ($n = 53$) and Group 2 ($n = 13$) according to the within-site dispersion: Group 1 with $\delta \leqslant 22°$ and Group 2 with $\delta \geqslant 37°$. Figure 3 shows the intensity decay curves of all specimens whose directions are shown in Fig. 2.

It is noted (Fig. 4) that the directions of all specimens in Group 2 (Fig. 4*a*) and those with $\Delta\Psi > 20°$ (Fig. 4*b* and 4*c*) moved towards the direction (275°, +05°) given by the Group 1 results. Since the minimum $\Delta\Psi$ is not always

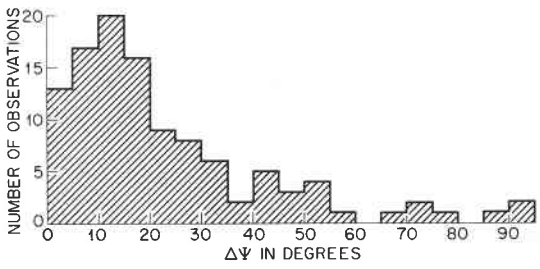


FIG. 5. Distribution of the index of within-specimen uniformity of direction given by the $\Delta\Psi$ equation (see text) and calculated for the 111 red-bed specimens. Specimens with a $\Delta\Psi \leqslant 20°$ have been used to calculate the red-bed pole (Table 1).

found at the same temperature, it is probable that the number of specimens falling in the category $\Delta\Psi \leqslant 20°$ and/or in Group 1 would have been larger if increments of 1 or 2 °C instead of 5 °C had been used. For example, at site 20 (Fig. 4*a*), the last measurement was obtained at 675 °C for three specimens and 670 °C for the other. The directions were evidently converging towards 275°, +05° or thereabouts. After cleaning at 680 and 675 °C respectively, no measurements could be obtained ($M_i < 10^{-7}$ e.m.u.). It is possible that cleaning at an intermediate temperature might have reduced the within-site dispersion enough for inclusion in the Group 1 data.

Nevertheless, there are enough samples in Group 1 (Table 1, Fig. 2) to show clearly that the Seal Group sediments acquired a magnetization in the direction 275°, +05°. Furthermore, the trend of the directional changes indicates that a component in the direction 275°, +05° is present in all other specimens (Fig. 3). The pole position obtained from this direction is 205° E, 05° N.

Chemical Experiment

With the aid of J. K. Park, 15 specimens were subjected to chemical leaching using the procedure described by Park (1970). The results were similar to those obtained by thermal demagnetization. After successively longer immersion times in hydrochloric acid, the direction moved progressively towards 275°, +05°. These sediments are well cemented, exhibit low porosity, and even after 1875 hr in acid, did not lose their red to pink color. Since it did not appear that, in this instance, further chemical leaching would contribute additional information and because the time required to leach them completely (possibly more than a year) was unknown, this experiment was abandoned. One example of chemical leaching is shown in Fig. 2*f*; the intensity dropped from 6.4×10^{-6} e.m.u. (NRM) to 1.8×10^{-6} e.m.u. (1875 h).

Time and Origin of Magnetization

All directions (Table 1, Figs. 2 and 4) are corrected for bedding. The beds at five sites (Group 1) dip ≃38° S, strike ≃092° and at the remaining nine sites dip ≃33° N, strike ≃083°. Since the mean direction of magnetization is roughly parallel to the strike of the beds, a conclusive fold test cannot be obtained. If no bedding tilt correction is applied, the mean direction (276°, +02°) is not significantly different and gives a pole at 204° E, 04° N; the precision K (=29) is smaller than the K (=56) obtained after correction for folding (Table 1). The small difference is not sufficient to prove statistically, but favors a pre-folding magnetization.

Stability tests show that no distinction can be made between the uniformly and the non-uniformly magnetized specimens; in both groups the magnetization is thermally very stable, and it appears that all specimens were magnetized by the same process. It is suggested that the non-uniformity (large $\Delta\Psi$) is caused by a field direction change (with respect to this area of the crust) *during* magnetization. The southeast direction (Figs. 2*f* and 4) indicates that a field reversal occurred at the time of magnetization, and most of the directions can be explained by supposing that the magnetization was acquired partly before and partly after the field reversal. For example, the direction of magnetization of specimen 32-1 (Fig. 2*f*) changes from east to west; the steep inclination (218°, −70°) obtained after cleaning at 670 °C corresponds to a minimum in the intensity decay curve (Fig. 3*f*) indicating that two oppositely directed components of magnetization, of about equal intensity, are then present. The similarity of the stability spectra of the east and west components suggests a continuing process of magnetization spanning time scales of field reversals. Such a lengthy process favors a magnetization of chemical origin. The chemical leaching (Fig. 2*f*) results also support this view. Combined chemical and thermal experiments on red beds (Roy and Park 1972) have shown that certain types of CRM (chemical remanent magnetization) are very stable thermally and are not completely demagnetized by cleaning at temperatures as high as 682 °C. Although the red beds studied were different (Pennsylvanian–Mississippian), the results in both cases are comparable. It is then suggested that most, if not all, of the magnetization of the Seal red beds is a CRM.

Possibility of Rotation

The results indicate that another event capable of producing a non-uniformly directed magnetization within a specimen may have taken place at the time of magnetization. There

is a high concentration of directions (Fig. 4) about 330° (and some about 150°) and low inclination. This and the fact that cleaning produces a streaking of directions towards 275°, +05° indicates that the magnetization may be composed of two components, one directed along 275° and one along ≃330°. Although it cannot be ruled out, it is improbable that the latter direction is attributable to a field directed along 330° and low inclination; this would require a paleopole at ≃150° E, 35° N which is well off the assumed polar path (relative to the Canadian Shield) for the Precambrian.

A plausible explanation is that the rock unit rotated about 55° clockwise during magnetization so that the streaking of directions is a record of a tectonic movement rather than a field direction change. Evidence for rotation is also provided by the Group 1 results where, upon demagnetization, many directions move along the low inclination circle in the fourth quadrant (a few examples are shown in Fig. 2). The study of the igneous rocks, which follows, also supports the hypothesis of a clockwise rotation of the Seal–Croteau area.

The red-bed results can be summarized as follows: part of the non-uniformly directed magnetization ($\Delta\Psi > 20°$) was acquired during the field reversal and/or rotation of the area while most of the magnetization of the Group 1 specimens was acquired after these two events.

Igneous Rocks (Croteau and Seal Volcanic and Seal Diabase)

All specimens were cored in the field and subjected to alternating field (a.f.) cleaning using two different a.f. demagnetizers (Larochelle and Black 1965, Roy *et al.* 1973). Three instruments were used for measuring: a biastatic magnetometer (Larochelle and Christie 1967), a SMI PAR spinner, and the astatic magnetometer mentioned previously.

In most specimens, three magnetic components were present: (1) a soft magnetization (M_1) which could be removed by a.f. cleaning in fields of 100 Oe—this magnetization is probably of viscous origin since its direction appears to be randomly distributed, (2) a magnetization (M_2) which is resistant to a.f. cleaning fields of 100–300 Oe—this magnetization is believed to be the initial magnetization acquired while the rocks were cooling; and (3) a magnetization (M_3) which is little affected by a.f. cleaning—this magnetization was probably acquired when hematite was formed or recrystallized. This view is supported by the resistance of M_3 to a.f. cleaning in fields up to 3000 Oe. Cleaning in these fields also indicated that in many specimens M_3 is directed *approximately* along the direction (275°, +05°) obtained from the sediments. These remarks are illustrated in Figs. 6, 7, and 8. After cleaning, there is a high concentration of directions about a mean 142°, +42° (Fig. 6*c*); the directions of samples containing both M_2 and M_3 ($\delta > 20$, Fig. 6*b*) are streaked towards 275°, +05°. Apparently M_1 which caused the direction to lie away from the great circle passing through the points 142°, +42° and 275°, +05° was removed leaving the resultant vector of $\mathbf{M}_2 + \mathbf{M}_3$.

As M_3 is rather clearly related to post-igneous hematization, this part of the study was limited to finding an estimate of the direction of M_2. This is the dominant component and probably formed at the time these igneous rocks originally crystallized. Since M_3 is more resistant to a.f. cleaning than M_2, the true direction of M_2 is not easily obtained in those rocks in which M_3 forms a significant proportion of the total magnetization. However, in general M_3 is much weaker than M_2 so that the best estimate of M_2 is found when the intensity ratio of M_2/M_3 is the largest (after removal of M_1). This ratio varies greatly from site to site and also from specimen to specimen within a site and its maximum occurs at different cleaning fields for different specimens. The decay curves (Fig. 8) are also extremely variable so that a blanket cleaning at a specific field would not provide adequate information. All specimens were cleaned in fields of 100, 150, 200, 250, 300, and (for some sites) 350 Oe and analyzed site by site after each cleaning. At a few sites, M_2 is completely dominated by M_3. The reason for this is megascopically obvious; for example, at site 12, the specimens can be described as hematitic basalt; the intensity is low and the magnetization remains unaffected by a.f. fields of 3000 Oe (Table 2). At other sites, M_3 appears to be absent. For instance, at site 7, the NRM directions are widely scattered

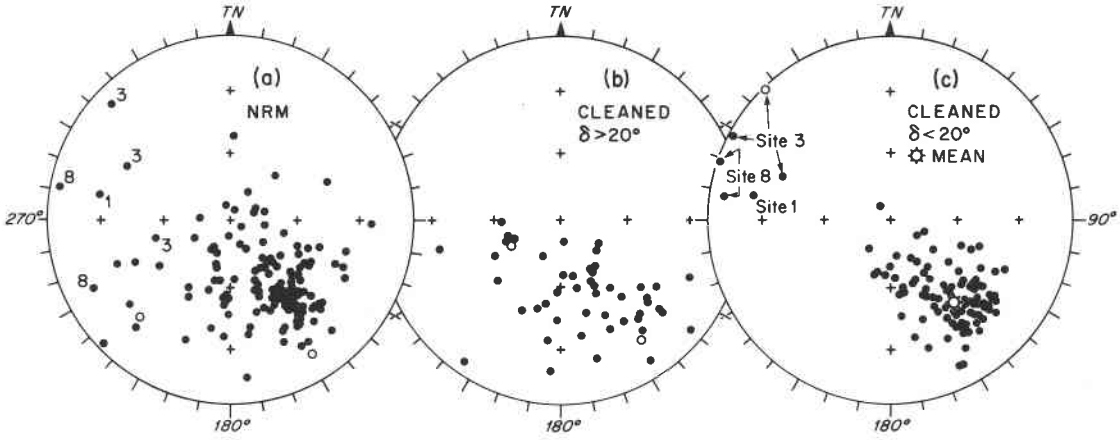


FIG. 6. Directions before and after a.f. cleaning of all cores obtained from igneous rocks. Equal area polar projection. (a) shows the directions of the resultant vector of three magnetizations: M_1, M_2 and M_3. (c) shows the directions of all cores used to calculate the mean direction (142°, +42°; Table 2) of M_2 which is probably the initial magnetization (see text). (b) shows the streaked directions towards 275°, +05° of the resultant vectors of $M_2 + M_3$. Closed symbols indicate positive inclinations.

($k = 4$) but group tightly ($k = 35$) about 143°, +42° after cleaning in a 250 Oe field. After cleaning in successively higher fields, there is no indication that M_3 is present; the scatter increases without any appreciable change in the mean direction and the intensities decrease (Fig. 8) until the directions are randomly distributed (field $\geqslant$ 1000 Oe). However, at most sites, both M_2 and M_3 are present. After cleaning in fields of 100–250 Oe, the within-site dispersion is reduced (Fig. 7*a*) or remains low (Fig. 7*c*) indicating the dominance of M_2; upon further cleaning in higher fields, the specimen directions often shift towards 275°, +05° (Fig. 7*e*) indicating the presence of M_3. In general, the direction displacement does not occur at the same cleaning field for all specimens from a site; furthermore the angular displacement varies from specimen to specimen (Fig. 7*d*). This has the effect of increasing the within-site dispersion. Therefore the minimum within-site dispersion indicates the cleaning field where M_3 has the least effect on the measured direction, that is, where M_2/M_3 is the largest.

Dispersions were calculated for each of the cleaning fields, and the mean site direction and intensity, relevant statistics, and the field at which the smallest dispersion was obtained are given in Table 2. Since the presence of M_3 tends to increase the within-site dispersion (Fig. 7*d*), its bias on the true direction of M_2 is further reduced by calculating the average of the site directions where the within-site dispersion $\leqslant$20°. The average direction (142°, +42°) thus obtained is considered a *rough* indicator of the initial magnetization (M_2). Mean site directions are plotted in Fig. 9. It appears that even in a few sites showing low angular dispersion (for example: 14, 33, and 34), M_3 may not be negligible. These site directions increase the steepness of the grand average and it is probable that, without the presence of M_3, the direction of M_2 would be less steeply inclined. However, the change would be small; exclusion of the three aberrant sites (14, 33, and 34) changes the average very little (140°, +38°). Therefore, although the bias of M_3 has not been entirely removed, the directional changes towards, and the tight grouping about 142°, +42° at many sites (Table 2, Figs. 7*a*, *b*, and *c*) leave little doubt that the initial magnetization was acquired roughly along a 142° declination with an inclination perhaps slightly less than +42° but probably at least +35°. The 142°, +42° (140°, +38°) direction gives a pole position at 152° E, 05° N (156° E, 07° N) which is some 50° away from the pole position obtained from the sediments.

The Group B results of Table 2 are for six

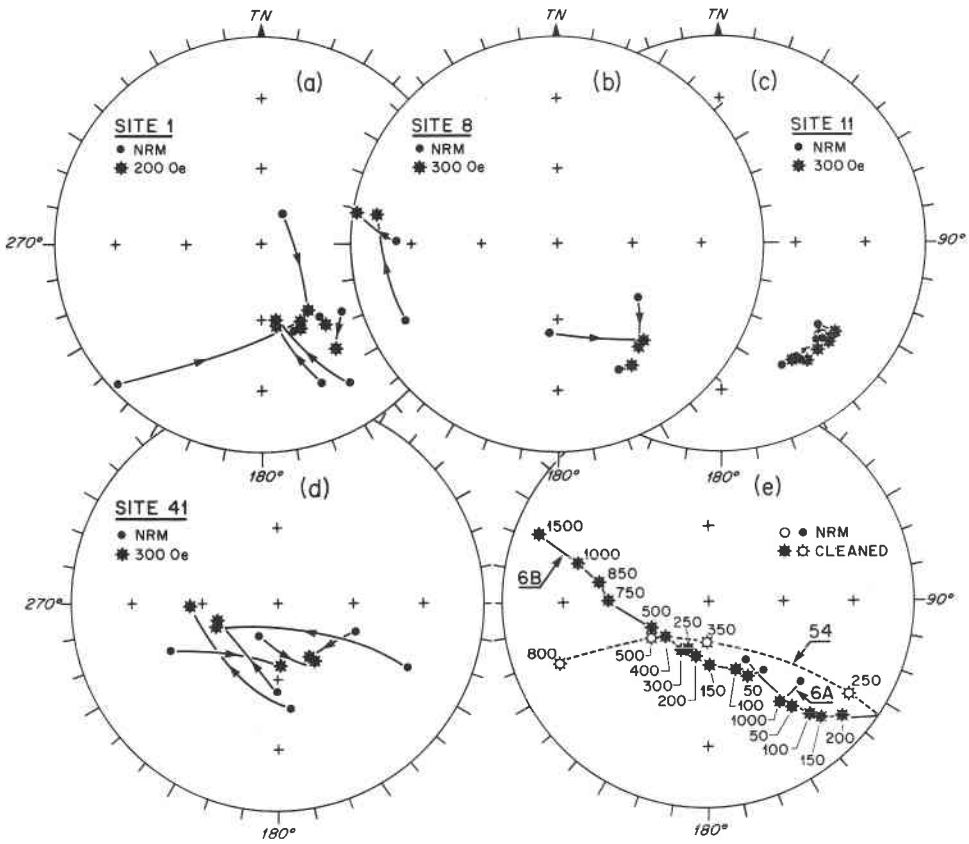


FIG. 7. Equal area polar projection. Directions before and after a.f. cleaning; (a) shows the removal of the randomly distributed M_1; the after cleaning direction shows a slight streaking between 142°+42° (M_2 direction) and ≃275°,+05°; (b) three directions group at 143°+36° (near M_2 direction); the other two directions clearly indicate the dominance of M_3; (c) only M_2 appears to be present; (d) shows two directions moving towards the M_2 direction while the four other directions are moving towards ≃275,+05°; this illustrates the variation of the ratio M_3/M_2 among specimens from the same site resulting in a large within-site dispersion when the intensity of M_3 is comparable to that of M_2; (e) direction displacements towards ≃275°,+05° which often occur in high a.f. cleaning fields. Specimen 6A, unlike 6B, remains near the M_2 direction until after 1000 Oe cleaning field; no stable magnetization was left after cleaning in higher fields.

samples that were excluded in calculating site-mean directions for accepted sites 1, 3, and 8. They were resistant to a.f. cleaning and undoubtedly reflect local, later, intense hematization of these samples and the development of a dominant M_3.

Chemistry of the Seal and Croteau Igneous Rocks

Partial chemical analyses of samples of igneous rocks collected at the paleomagnetic sampling sites are given in Table 3. Variation in chemistry among the rocks of the Seal sites is small and the means are probably quite representative of the chemistry of the section which was sampled. The volcanics and diabase are chemically similar, suggesting a common magma source. They differ in the greater degree of oxidation of the volcanics as compared with the intrusions, and the significantly higher calcium content of the diabases. The oxidation of the volcanics is attributed to their extrusion in a continental environment. The lower calcium content of the Seal Group volcanics is probably

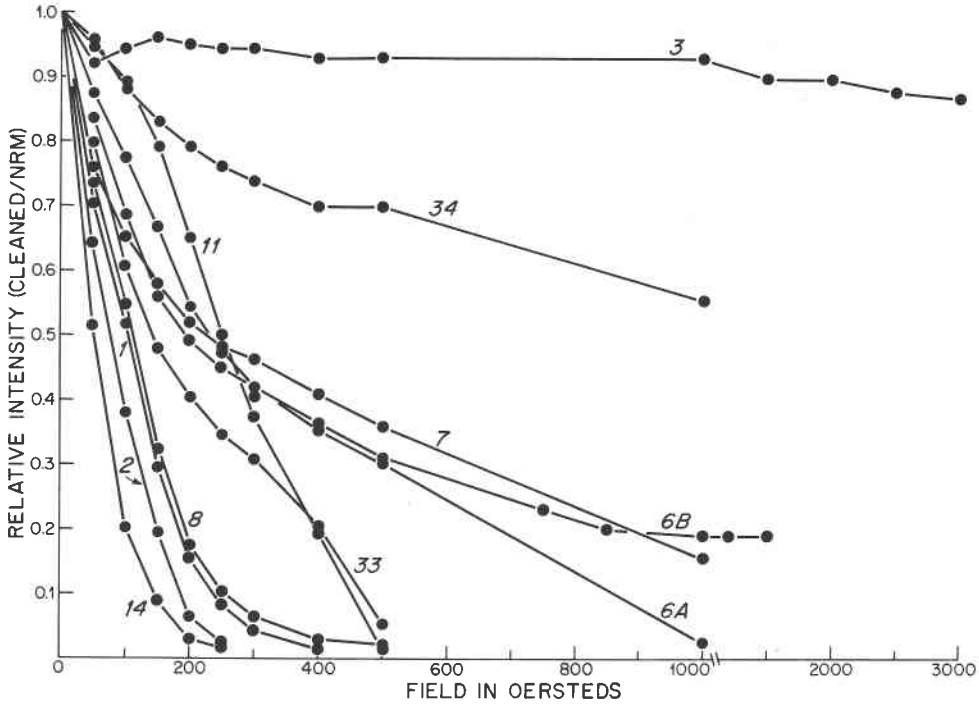


FIG. 8. Normalized intensity decay curves of different specimens showing the wide variation amongst specimens.

due to the leaching affects that accompanied the oxidizing processes (Cameron and Baumann 1972). Most of the volcanic rocks sampled for paleomagnetic work are undersaturated tholeiites with both olivine and orthopyroxene in their norms. Some analyses of volcanics from lower in the Seal sequence (not given in the table) are distinctly alkalic and their norms contain significant quantities of nepheline. Because they are less altered the diabase sills probably characterize the parent magma better than do the volcanics. One third of the sills sampled exhibit a minor amount of nepheline in their norms.

The Croteau Group volcanics sampled in this work show greater chemical variation than the Seal Group volcanics. They are clearly more salic and are andesites or trachyandesites. They are also rather highly oxidized, presumably because of subaerial extrusion.

Magnetization Direction of M_2 and M_3

The igneous rocks (volcanics and diabase intrusions) of the Croteau and Seal Groups give evidence of two stable magnetic components. The stronger of these is directed at a moderate angle to the southeast; the weaker at a shallow angle to the west. The weaker westerly-directed magnetization appears to be the later one because it is clearly related to the presence of hematite, throughout the rocks of some sites or within individual cores of less homogeneous sites. The hematitic nature of specimens carrying this component is indicated by redness, by a lower ferrous/ferric ratio in the rock as a whole (Table 3) and by the extreme resistance of these rocks to a.f. demagnetization. In the volcanic rocks the hematization clearly postdates the consolidation of the flows as its presence is controlled by columnar joints, irregular fractures, or breccia zones. Similar evidence of hematization is almost invariably reported in volcanic rocks of subaerial origin. In such rocks the hematization is generally thought to be very early, perhaps related to the circulation of hot, oxygen-charged ground waters during the cooling of the flows. One would expect that the hypabys-

TABLE 2. Analysis of intensities and directions of volcanics and diabases before (M_n) and after (M_F) cleaning in the alternating field indicated

			M_n(NRM)				Peak	M_F (cleaned)					Pole position
Site	N	n	M_i	$D°, I°$	R	$\delta°$	field	M_i	$D°, I°$	R	K	$\delta°$	°E,°N
							Group A						
							Seal volcanics						
1*	8	7	477	158,+48	5.659	36	200	67	153,+53	6.912	69	9	
3*	6	3	12	151,+44	2.952	10	3000	13	152,+38	2.952	41	10	
5	4	4	564	167,+45	3.694	23	250	48	170,+41	3.539	7	28	
6*	4	4	670	129,+43	3.952	9	250	246	140,+43	3.915	35	12	
7*	5	5	1475	174,+38	3.924	38	250	158	143,+42	4.884	35	12	
8*	5	3	663	153,+48	2.851	18	300	57	143,+36	2.992	241	4	
12	4	4	5	201,+51	3.607	25	3000	4	204,+48	3.493	6	29	
Mean		5		153,+45	4.905	11			146,+43	4.963	107	7	149,06
													(6,10)†
							Seal diabases						
2*	6	6	1975	113,+56	5.709	18	100	1132	125,+41	5.978	223	5	
11*	6	6	33	141,+34	5.952	7	300	12	138,+31	5.968	154	6	
9	3	3	2682	153,+80	1.841	52	200	264	147,+09	1.560	1	59	
13	6	6	2888	191,+30	5.673	19	150	856	183,+61	5.610	13	21	
14*	3	3	1457	220,+66	2.841	18	100	332	172,+59	2.953	43	10	
15*	6	6	3211	154,+74	5.691	18	100	1644	158,+46	5.910	55	10	
50*	6	6	1162	145,+42	5.980	5	150	374	148,+24	5.958	118	7	
51*	6	6	2315	148,+46	5.788	15	200	350	144,+39	5.739	19	17	
52*	6	6	1683	141,+43	5.982	5	250	290	137,+37	5.942	86	8	
53*	6	6	932	138,+36	5.909	10	200	255	137,+22	5.640	14	20	
54*	6	6	463	144,+52	5.982	4	200	120	141,+30	5.949	98	7	
55	6	6	1199	151,+52	5.940	8	100	518	160,+32	5.550	11	22	
													(7,12)†
Mean		9		141,+54	8.238	24			143,+37	8.726	29	14	153,08
Seal average		14		148,+50	13.395	17			144,+39	13.672	40	12	152,08
													(5,8)†
							Croteau volcanics						
33*	5	5	32	094,+75	4.825	15	200	6	156,+73	4.827	23	15	
34*	6	6	8	173,+71	5.891	11	1000	5	159,+65	5.773	22	16	
35*	5	5	1295	174,+64	4.065	36	200	60	141,+45	4.763	17	18	
36	5	5	33	186,+58	3.640	43	250	10	177,+50	4.320	6	30	
37	5	5	163	168,+58	4.756	18	200	47	149,+72	4.494	8	26	
39*	6	6	193	131,+68	5.870	12	150	33	119,+37	5.882	43	11	
40*	6	6	1304	143,+52	5.416	25	150	99	127,+39	5.862	36	12	
41	6	6	2000	162,+61	5.161	31	300	66	211,+72	5.592	12	21	
Mean		5		147,+68	4.875	13			136,+52	4.791	19	16	154,−05
													(16,26)†
Croteau +													
Seal average		19		148,+54	18.082	18			142,+42	18.351	28	15	152,05
													(5,8)†
							Group B‡						
1		1	151	290,+15	—	—	200	162	290,+24	—	—	—	
3		3	18	295,+36	2.707	26	3000	17	302,+12	2.823	11	20	
8		2	62	262,+12	1.903	—	300	61	284,+04	1.987	—	—	
Average		3		282,+22	2.865	17			292,+13	2.944	36	11	194,+18

NOTES: N = number of specimens measured; n = number of specimens used; M_i = intensity × 10^{-6} e.m.u. (CGS system); R = resultant vector; K = precision estimate; δ^0 = angular standard deviation.
*Sites (where $\delta < 20°$) used to obtain means and averages.
†Semi-axes (δp, δm) of the ellipse of confidence at $P = 0.05$.
‡This group is composed of the six samples that were excluded from Group A.

sal diabase sills of the Seal Group would be less exposed to such alteration and would exhibit less evidence of hematization. Such is the case. We conclude from the above evidence that the flat, westerly-directed magnetic component in the igneous rocks is the result of

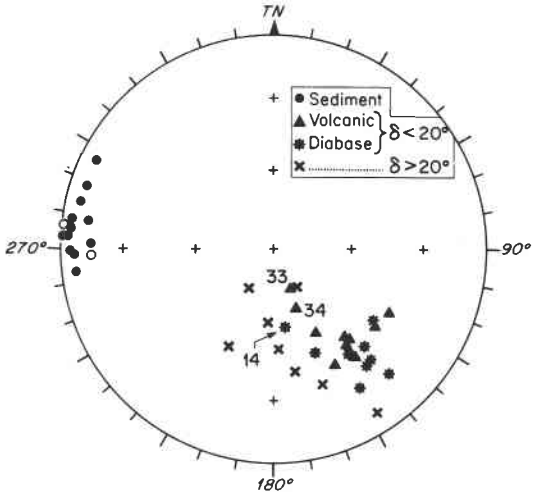


FIG. 9. Mean site directions of Group 1 sedimentary and Group A igneous rocks. Crosses indicate the directions (all positive inclinations) of the sites which were not used for calculating the averages. The igneous rocks directions show a concentration about 142°,+42° and streaking along the great circle passing through 142°, 42° and the direction (175° + 05°) obtained from the sediments.

secondary hematization and is therefore a CRM. Comparison of directions between M_3 and the red-bed CRM also support this view and may provide some information about the time of hematization. In some specimens, the M_3 direction (Fig. 7b and e) is clearly comparable to the Group 1 direction indicating that the two magnetizations were probably acquired in the same field and perhaps at the same time; in addition, at some sites, the streaking of directions towards the west (Fig. 7*d*) is similar to the one observed in the red beds, and which was attributed to the field reversal. So, it may be that the hematization occurred at the time of magnetization of the red beds.

The strong, southeasterly dipping magnetization of the Seal–Croteau igneous rocks is thought to have originated at the time these rocks formed and is probably a TRM. It has a direction which is almost identical to that (138°, +31°) of the Michael Gabbro (Fahrig and Larochelle 1972) and the Croteau and Michael rocks have similar Rb/Sr ages so altogether these rocks containing a similar southeasterly magnetization extend through an area of several tens of thousands of square kilometers. This militates against the possibility that the southeasterly component is a later one systematically imposed throughout this area on rocks whose mineralogy varies greatly. Furthermore the Michael Gabbro shows no evidence of post-crystallization alteration and the Seal–Croteau rocks show only evidence of a late CRM (hematization), the effects of which have been greatly reduced. It is noteworthy in this regard that the Seal diabase which is least hematized exhibits a magnetization (143°, +37°) that most closely approaches that of the Michael Gabbro.

Discussion

The shift in pole position (50° of arc) between the development of the igneous (TRM) and the later hematite (CRM) and the polarity reversal between the development of these magnetizations was a surprising element in this study. As noted above, the hematization of subaerial volcanic sequences is generally described as a late magmatic event. In the present case the hematization or the recrystallization of earlier hematite appears to have taken place significantly later than the original igneous activity. Two things suggest this, the very large shift of the paleomagnetic pole relative to this area of crust during the interval between the development of the TRM and the CRM, and the polarity reversal. The Croteau volcanics form a thick sequence that appears to be unconformably overlain, at least locally, by the Seal Group, and the igneous rocks of these two groups as well as the Michael Gabbro appear to exhibit one primary polarity. As these rock sequences are likely to have formed over several tens of millions of years, it is suggested that polarity reversals were widely spaced during this part of the Helikian and that the two CRM's (M_3 and the red beds) developed millions of years after the initial igneous activity.

Construction of Apparent Polar Wandering Paths

Many paleomagnetic data of high quality are now available from Precambrian rocks of various ages and localities from within the Canadian Shield so that construction of Precambrian polar wandering paths and of strati-

TABLE 3. Chemical analyses of the Seal and Croteau igneous rocks

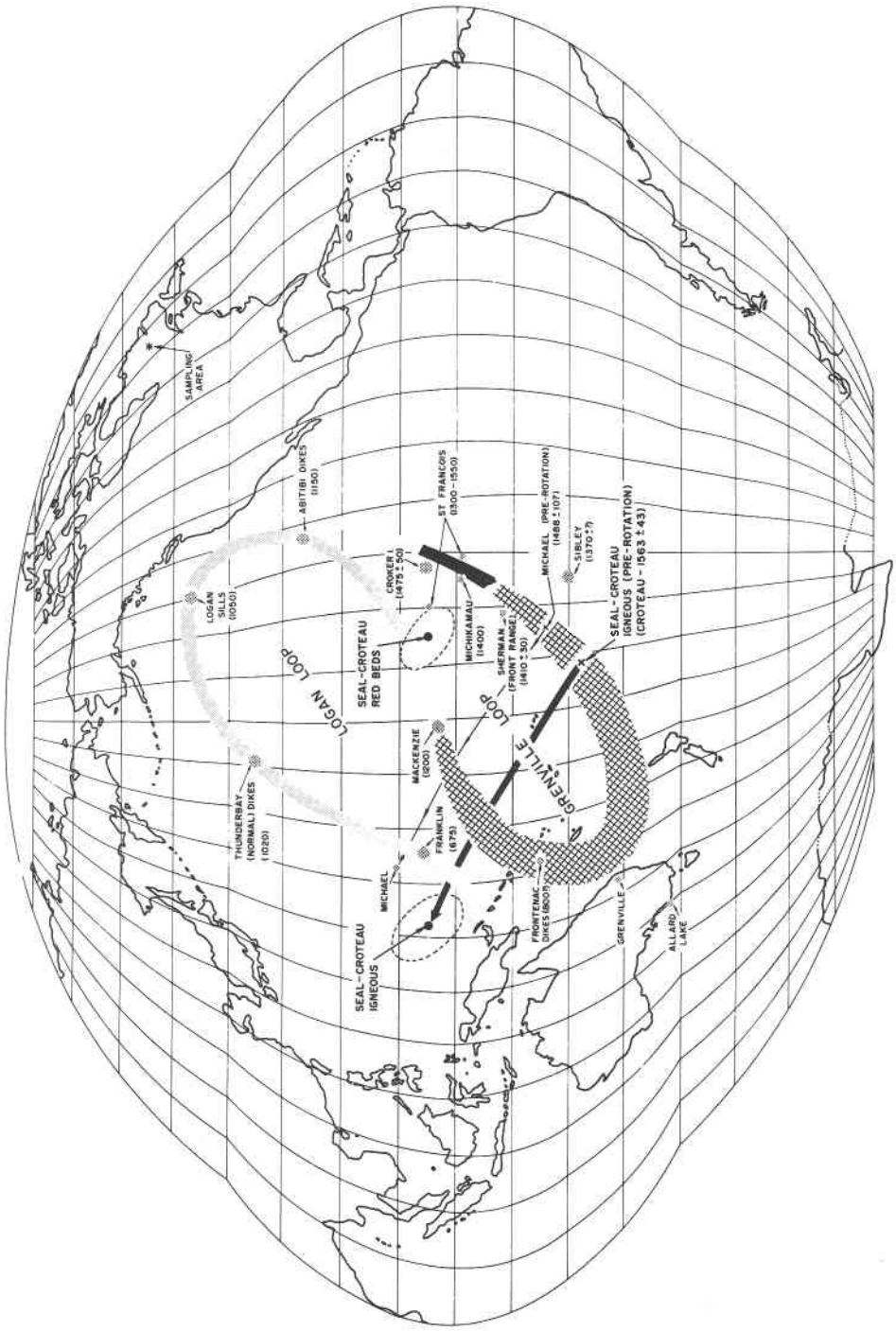
Site no.	SiO_2	Al_2O_3	Fe_2O_3	FeO	MgO	CaO	Na_2O	K_2O	TiO_2	MnO	P_2O_5	FeO/Fe_2O_3
						Seal volcanics						
1	44.6	15.6	3.72	9.44	8.1	7.9	2.8	0.5	2.04	0.17	0.23	2.5
3	48.1	16.2	8.35	4.55	6.9	3.6	3.9	1.6	2.37	0.24	0.43	.54
5	48.1	16.0	4.91	7.43	5.0	6.2	4.1	1.5	2.28	0.21	0.47	1.5
6	44.2	16.8	8.68	6.42	7.3	5.1	3.1	0.9	2.28	0.21	0.24	.74
7	43.3	15.7	7.84	6.35	7.4	6.1	3.5	0.4	2.26	0.20	0.25	.81
8	46.1	14.8	7.47	6.57	6.1	4.6	5.3	0.4	2.21	0.20	0.44	.88
12	42.6	16.2	7.47	2.26	7.0	7.5	5.1	0.2	1.54	0.19	0.16	.30
Mean	45.3	16.0	6.92	6.14	6.8	5.8	4.0	0.8	2.14	0.20	0.31	1.04
						Seal diabase						
2	47.1	16.6	2.92	9.73	6.5	9.2	3.0	0.8	1.93	0.17	0.26	3.3
9	46.2	16.5	4.35	8.51	6.5	8.6	2.9	0.9	1.97	0.18	0.27	2.0
11	46.8	17.8	3.24	9.87	6.4	9.7	3.0	0.1	1.74	0.19	0.18	3.0
13	48.0	14.7	3.68	9.76	5.7	11.2	3.1	0.7	2.56	0.22	0.22	2.6
14	47.6	14.5	4.13	10.37	4.6	9.8	3.6	0.8	2.70	0.23	0.28	2.5
15*	45.0	10.3	6.91	12.96	4.8	10.8	2.5	0.9	5.11+	0.27	0.53	1.9
50	45.7	16.5	2.68	9.94	7.0	9.1	3.1	0.8	1.92	0.18	0.28	3.7
51	47.3	16.0	3.41	10.37	6.2	8.6	3.3	0.9	2.32	0.19	0.33	3.0
52	46.4	16.5	2.76	9.87	6.7	9.1	3.1	0.8	2.01	0.17	0.30	3.6
53	45.8	15.9	2.07	10.80	7.4	9.3	3.0	0.7	1.96	0.17	0.26	5.2
54	46.1	16.3	9.23	5.20	7.4	4.9	4.0	0.9	2.23	0.22	0.46	.56
Mean	46.7	16.1	3.80	9.44	6.4	9.0	3.2	0.7	2.15	0.19	0.28	3.0
						Croteau volcanics						
33	51.8	13.6	1.46	3.69	5.8	5.8	3.7	3.0	0.64	0.12	0.22	2.5
34	62.7	16.3	3.80	0.46	1.0	2.9	3.8	5.0	0.82	0.10	0.31	.12
35	51.5	18.8	5.70	2.70	2.6	7.7	3.6	1.7	1.15	0.11	0.18	.47
36	49.0	16.5	5.96	3.19	6.2	8.2	2.8	0.9	1.19	0.16	0.17	.54
37	50.9	17.6	7.07	2.69	4.1	7.4	4.1	0.5	1.68	0.18	0.28	.38
39†	61.3	16.9	2.04	1.18	1.3	2.6	4.2	3.8	0.41	0.09	0.08	.58
40	50.6	17.9	5.48	4.55	3.7	8.4	3.4	1.7	1.34	0.15	0.23	.83
41	51.6	18.8	5.24	4.77	4.0	7.7	3.1	1.8	1.44	0.15	0.24	.91
Mean	52.6	17.1	4.96	2.76	3.9	6.9	3.5	2.1	1.18	0.14	0.23	.56

NOTES: Analyses by Rapid Methods Group, Analytical Laboratories, Geological Survey of Canada. SiO_2, Al_2O_3, MgO, CaO, K_2O, TiO_2, MnO by X-ray fluorescence analyses; Na_2O, P_2O_5 by rapid chemical methods; ferric and ferrous iron determinations by the classical analytical method (Pratt).

*This analysis is of rock from the upper part of a thick diabase sill which exhibits iron and titanium enrichment. The values for this rock are not included in the mean for the Seal diabase.

†This analysis is of a fine- to medium-grained volcanic tuff. It has not been included in the mean for the Croteau volcanics.

graphic charts should no longer be an exercise in futility. In its simplest form a polar wandering path relates only to an area of rock within which there has been either no relative movements during the time represented by the path, or within which relative movements can be independently determined. The Logan Loop (Robertson and Fahrig 1971) for example which begins with the Mackenzie interval and ends with the Franklin was constructed from key pole positions obtained from rocks of Helikian and Hadrynian age. All the rocks yielding the key points are represented within the Superior Orogen and none of the rock units is thought to have been disturbed relative to the others since it became stably magnetized. When polar wandering paths are constructed by incorporating data from rocks within the Superior as well as from rocks a considerable distance away from the Superior Orogen (*e.g.* Spall 1971), it is then implied that all of these rocks belong to a unit which has remained intact since the magnetization was acquired. We believe that, at the present time, this is an unwarranted assumption. We propose that the Superior Orogen be considered as the 'reference block' in paleomagnetic work involving the Precambrian of North America. This point is emphasized on Fig. 10 by showing pole positions from rocks well away from the 'Superior reference block' with smaller symbols. Pole positions from rocks within the Superior are


SAMPLING AREA
ABITIBI DIKES (1150)
LOGAN SILLS (1050)
LOGAN LOOP
THUNDERBAY (NORMAL) DIKES (1020)
SEAL-CROTEAU RED BEDS
CROKER I (1479 ± 50)
ST FRANCOIS (1500-1550)
MICHIKAMAU (1400)
SHERMAN (PRONT RANGE) (1410 ± 50)
MICHAEL (PRE-ROTATION) (1488 ± 107)
SIBLEY (1370 ± ?)
SEAL-CROTEAU IGNEOUS (PRE-ROTATION) (CROTEAU - 1563 ± 43)
GRENVILLE LOOP
MACKENZIE (2000)
FRANKLIN (675)
MICHAEL
FRONTENAC DIKES (1600)
SEAL-CROTEAU IGNEOUS
GRENVILLE
ALLARD LAKE

considered to be key reference points for comparison of poles from rocks of similar age outside the Superior. The key poles for rocks similar in age to the Michikamau anorthosite, for example, may be considered to be those for the Sibley (Robertson 1973) and Croker Island complex (Palmer 1969) which lie on the edge of the Superior. This concept is used in this discussion of the Precambrian polar path.

Evidence for Rotation

The Croteau and Seal Groups lie within the confines of the Grenville Structural Province as defined by Stockwell (1964) and the Grenvillian was the last orogeny that significantly affected most of these rocks. We believe, however, that the magnetization direction derived from the igneous rocks of these groups dates from the time of their original consolidation (about 1500 m.y. ago) and when corrected for dip gives the direction of the earth's magnetic field relative to these rocks at that time. The paleomagnetic pole for the igneous rocks, however, lies about 65° of arc away from the pole position for rocks of similar age in the Superior Province.

Fahrig and Larochelle (1972) have suggested that the block containing the Michael Gabbro (mean location: 301° E, 54.5° N), which has an age and pole position similar to that of the igneous Seal–Croteau, rotated about a local vertical axis. The results obtained, in this study, from both the red beds and the igneous rocks evidently support such a rotation and indicate that the Seal–Croteau area was most probably part of the same rotated block. At the present time, the best *estimate* of the amount of rotation is obtained from the red-bed results and appears to be about 55° of arc. Figure 10 shows the 'pre-rotation' Seal–Croteau and the Michael poles obtained by applying that correction. It is clear that correction for rotation brings these poles much nearer the other poles of similar age. When compared to the poles obtained from units within the Superior, it is found that they are compatible with the Sibley ($\delta p = 7°$; $\delta m = 13°$) taking into account the limits of confidence and the uncertainty in determining both the angle of rota-

FIG. 10. The circular symbols are pole positions for various Precambrian rock units. The larger symbols are poles for rocks either within the Superior structural province or close enough to its border to suggest that the rocks have not moved relative to the Superior province since they acquired their stable remanent magnetization. Figures in brackets near the poles are radioisotopic ages, the more reliable ages (Rb/Sr) are indicated with confidence limits and are calculated using $\lambda = 1.39 \times 10^{-11} \times y^{-1}$, the remainder are K–Ar ages and are probably minimum ages. References for the poles plotted here are — *Croker Island*, Palmer (1969), *St. François*, Hays and Scharon (1966) and Hsu *et al.* (1966), *Michikamau*, Murthy *et al.* (1968), age ±80 m.y., *Sherman Front Range*, Eggler and Larson (1968), age from Ferris and Krueger (1964), *Michael*, Fahrig and Larochelle (1972), *Sibley*, Robertson (1973), *Seal–Croteau igneous*, Roy and Fahrig (this study), *Allard Lake*, Hargraves and Burt (1967), *Grenville (diorites)*, Irving *et al.* (1972*c*), *Frontenac dikes*, Park and Irving (1972), *Mackenzie*, Fahrig *et al.* (1965), Fahrig and Jones (1969), Robertson (1969), Irving *et al.* (1972*b*), *Seal–Croteau red beds*, Roy and Fahrig (this study), *Abitibi dikes*, Fahrig *et al.* (1965), Larochelle (1966), *Logan sills*, Dubois (1962), Palmer (1970), Robertson and Fahrig (1971), *Thunderbay (normal) dikes*, Palmer (1970), Robertson and Fahrig (1971), *Franklin*, Fahrig *et al.* (1971), Robertson and Baragar (1972), Fahrig and Schwartz (1973).

The proposed Seal–Croteau and Michael 'pre-rotation' pole positions have been obtained by assuming that, after the initial magnetization of the igneous rocks, the block containing these rock units rotated clockwise about a local axis by 55° of arc as suggested by the red-bed results. Smaller angles of rotation would locate the poles along the drawn lines. The Frontenac dikes, the Grenville (diorites), and the Allard Lake poles may well derive from magnetizations that "see through" the K–Ar event (Irving *et al.* 1972*c*). It is suggested that these pole positions may be due to polar wandering and belong to the Grenville Loop. The Allard Lake has not been given great weight because it has a large circle of confidence. The Grenville Loop is shown here as a wide cross-hatched band to emphasize the uncertainties involved in its exact location. It is proposed that the excursion of the pole into the southern hemisphere be named the Grenville Loop as the possible existence of this loop has been derived from Grenville rocks. It is suggested that the red-bed pole possibly belongs to the eastern limb of the Logan Loop (Robertson and Fahrig 1971).

tion of the block and, as discussed previously, the inclination of M_2: a (possible) lower ($\leqslant 7°$) inclination of M_2 would produce a pole nearer the Michael and Sibley poles, thereby improving the agreement between these pole positions. The difference between the Seal–Croteau (and Michael) and the Croker Island poles may be attributable to a relative displacement of the two locations or alternately to polar wandering. Since there is no evidence that the Sibley Formation and the Croker Island Complex have moved relative to one another, the agreement between the Seal–Croteau and Sibley poles leads to the suggestion that the difference is attributable to polar wandering. It is noted that the difference may not be as large as shown in Fig. 10 because the Croker pole may be slightly too far north owing to the incomplete removal of a steeply inclined secondary magnetization (see Palmer (1969)).

The Grenville Front Problem

Because the Seal and Croteau Groups are situated near the Grenville Front, the results have a bearing on the suggestion (Irving *et al.* 1972*c*) that the Grenville Province was once thousands of kilometers away from the remainder of the Canadian Shield. If so, is the Grenville Front a plate juncture? Large geological features such as folded rocks of the Labrador Geosyncline and the great belt of Helikian anorthosite intrusions which cut across the Grenville Front (Stockwell *et al.* 1970, p. 121) make it improbable that the rocks on each side of the Front originated from two separate land masses. On a more local scale, the likeness of gently folded correlatives of the Seal Group, which form the Shipiskan Plateau (just northwest of Fig. 1), to the folded Seal strata suggests that they were not formed thousands of kilometers away from each other. Likewise, the paleomagnetic results do not indicate that the Seal–Croteau–Michael block was formed at such a great distance from the Superior Orogen. So, according to the geological and paleomagnetic evidence, it appears that the Grenville Front itself is not a plate juncture. However, it is noted that the localities used by Irving *et al.* (1972*c*) were all a good distance south of the front so that the possibility (Donaldson and Irving 1972, Donaldson *et al.* 1973) remains that a plate juncture is located at some distance south of the Front as described by Stockwell *et al.* (1970, p. 47)—a possibility which has been implied by Irving *et al.* (1972*c*, p. 346) when they provide an explanation for the various geological features found immediately south of the Front.

A Proposed Grenville Loop

Recent reconstructions (Spall 1971, Robertson and Fahrig 1971, Irving *et al.* 1972*a*) of the Precambrian polar wandering path for North America show large polar shifts from 2000 to 1500 m.y. ago and from 1200 to 600 m.y. ago with an almost stationary pole near the equator from 1500 to 1200 m.y. ago. Figure 10 shows three reasonably well-dated poles in the southern hemisphere. We suggest that, during an interval of time which began before the formation of the Seal, Croteau, and Michael rocks (say 1550 m.y.) and ended at the time of the MacKenzie igneous events (about 1200 m.y.), the pole undertook an excursion (the Grenville Loop) into the southern hemisphere. As suggested by Irving *et al.* (1972*c*), the ages of magnetization of the Allard Lake, the Grenville (diorites), and the Frontenac dikes may be much older than their apparent K–Ar ages; however, as an alternative solution to the displacement of the southern part of the Grenville Province, we suggest that these poles may be a reflection of polar wandering and may contribute to the definition of the loop. Future results from the Grenville should clarify the present uncertainty.

Rocks from the Superior Province that formed in the interval 1550–1200 m.y. ago should yield poles that lie well south of the present-day equator and evidently *such poles are required to define more accurately the 'reference' path* of the Grenville Loop.

The extent of the block involved in the proposed 'local' rotation has still to be established but Fahrig and Schwarz (1973) suggest that at least the entire northeastern part of the Grenville Province was involved. The tectonic effects of such a rotation should be geologically visible. One would expect the development of great curved zones of crushed and mylonitized rock as the irregular edge of one block moved against that of another. One such zone may be present in the Shabogamo

map area south of the Labrador Trough (Fahrig 1960); a second may be the broad zone of shearing described by Stevenson (1970) south of Benedict Mountains near the Labrador coast. The clockwise rotation of the tectonic elements of the Kaniapiskau Supergroup within the Grenville Province may also be a result of this rotation.

Time of Magnetization of the CRM

The pole positions obtained from the red beds and the igneous rocks may be used to set limits with regard to the probable age of the CRM. Since the CRM was acquired after the time of the primary magnetization (M_2) of the igneous rocks, it is younger than 1563 m.y. ago, and its pole should be found along the polar path at a more recent point in time. Because of its position (Fig. 10), the red-bed pole could be related in age to the lower part of the eastern limb of the Logan Loop (Robertson and Fahrig 1971). The post-rotation CRM (Group 1 results) may therefore be as much as, but probably not more than, 400 m.y. younger than the primary magnetization of the igneous rocks. The rotation that occurred during the early stage of magnetization (of the red-bed CRM) may then have taken place during early Grenvillian orogenic and thermal (hydrothermal) events between 1200 and 1500 m.y. ago. This agrees with K–Ar ages obtained on Seal rocks which have a maximum age of 1045 m.y. (Wanless 1970) and which may reflect the last thermal effects of the Grenvillian in this area. It is noted that the red-bed results also indicate that the rotation and a field reversal occurred at about the same time so that, as more data become available, precise determination of the time of rotation would provide a polarity marker for dating purposes, or vice versa, timing of the reversal would determine more accurately the rotation time.

Tectonic Implications

Absolute age determinations, especially of Precambrian rocks, are difficult to obtain and their relationship to paleomagnetic studies has been discussed by a number of authors (*e.g.* Spall 1971, Fahrig and Larochelle 1972). In Fig. 10, the best estimate of age for different paleomagnetic studies are given, some of them with their confidence limits. It may be pointed out that the samples used for age determination were not necessarily in the vicinity of the rocks studied paleomagnetically. It then appears that the polar path is best reconstructed by drawing the path through *reliable* pole positions although this may sometimes conflict with certain age determinations. Reliable pole positions (after correction for tectonic movement) should fall onto or near the path and their relative position along this path should give their relative age of magnetization. Thus, it is proposed that the polar path (Fig. 10) provide the relative ages of different rock units and can be used to describe, for example, a sequence of events in the Labrador area near the Grenville Front.

The paleomagnetic data and geological observations support the following points.

(1) The Croteau, Seal, and Michael rocks postdate the intrusion of the Michikamau anorthosite: a relationship already suggested by the fact that Seal strata appear to unconformably overlie the Harp anorthosite.

(2) The rift system in which the supracrustals were deposited developed after the intrusion of the Harp and Michikamau anorthosites.

(3) The Seal–Croteau and Sibley rocks may have been deposited in the same rift system which extended towards the interior of the continent.

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